

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
CODE: CSA0713

COURSE

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I _M.Greeshma (192524195)_ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Greeshma

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Sol..

1

School of Computing (/25)

Subnetting the first half of the /24 block.

- **Network Address: 192.168.100.0**
- **Broadcast Address: 192.168.100.127**
- **Valid Host Range: 192.168.100.1 – 192.168.100.126**
- **Usable Hosts: 126**

2

School of Electronics (/26)

Subnetting the next quarter of the block.

- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range: 192.168.100.129 – 192.168.100.190
- Usable Hosts: 62

3

School of Mechanical Engineering (/27)

Subnetting the next eighth of the block.

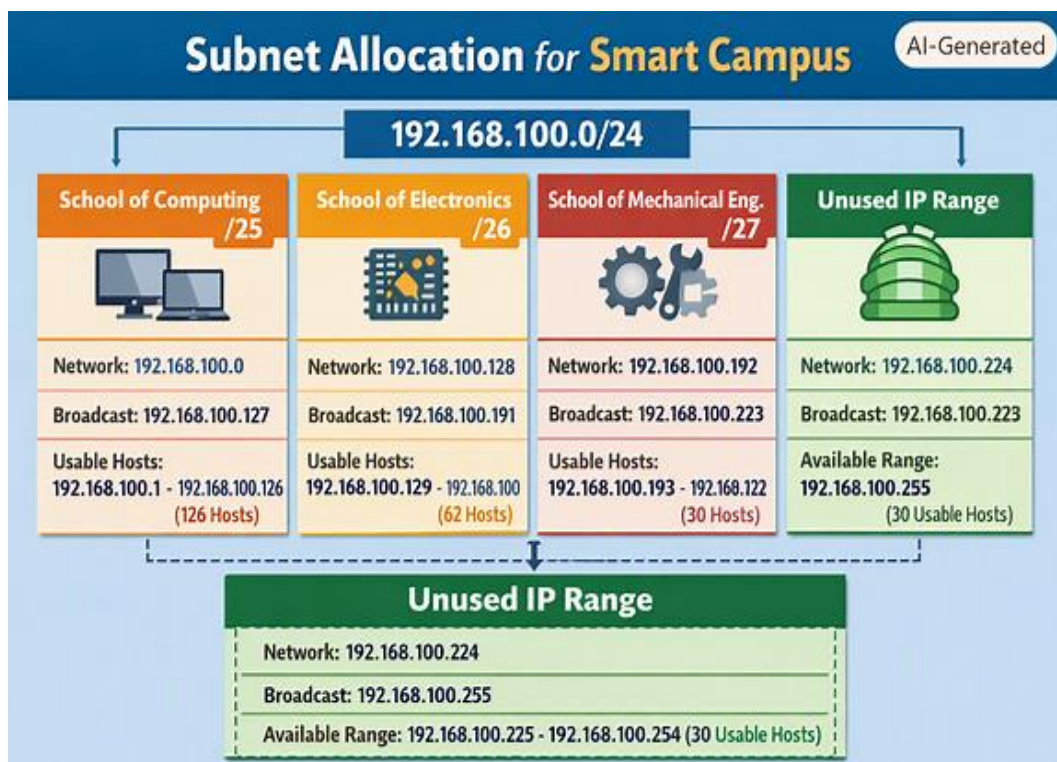
- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222
- Usable Hosts: 30

4

Remaining Unused Range

The leftover addresses after assigned subnets.

- Unused Range: 192.168.100.224 – 192.168.100.255
- This covers a /27 (192.168.100.224/27) and a /28 (192.168.100.240/28)
- Can be allocated to future departments or services



2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**
- 3. Calculate the number of usable hosts in each subnet.**
- 4. Identify the unused IP address range remaining within the allocated IP block.**

Sol..

□Network Address and Broadcast Address

Department	Subnet Mask	Network Address	Broadcast Address
Software Development	/24	10.10.0.0	10.10.0.255
Quality Assurance	/25	10.10.1.0	10.10.1.127
Human Resources	/26	10.10.1.128	10.10.1.191
Executive Management	/27	10.10.1.192	10.10.1.223

3Valid Host IP Address Range

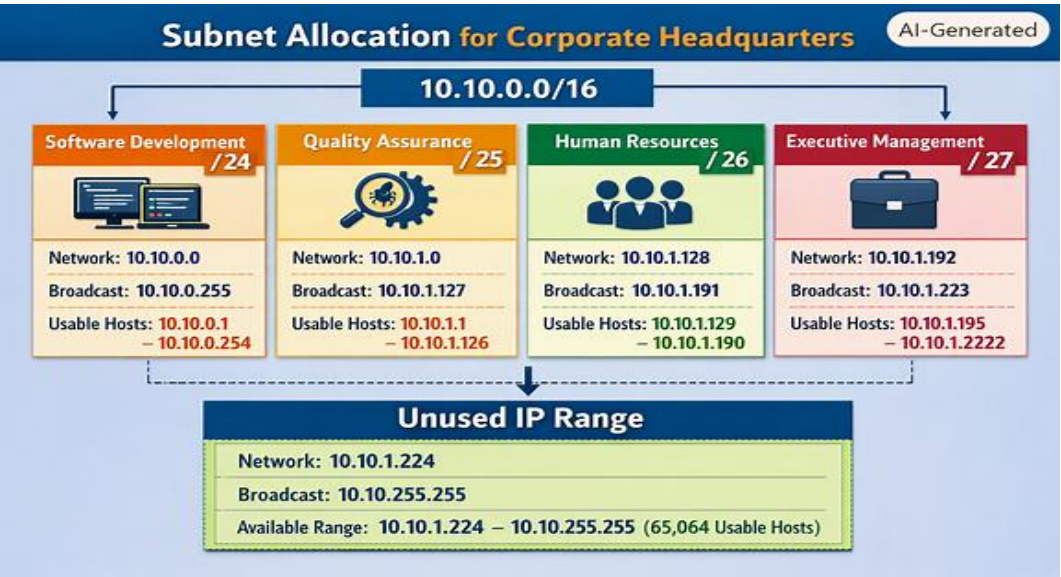
Department	Valid Host Range
Software Development	10.10.0.1 – 10.10.0.254
Quality Assurance	10.10.1.1 – 10.10.1.126
Human Resources	10.10.1.129 – 10.10.1.190
Executive Management	10.10.1.193 – 10.10.1.222

3Number of Usable Host Addresses

Department	Usable Hosts
Software Development	254
Quality Assurance	126
Human Resources	62
Executive Management	30

4Remaining Unused IP Address Range

- Unused Range: 10.10.1.224 – 10.10.255.255
- Total Unused Usable Hosts: $65,536 - (254 + 126 + 62 + 30) = 65,064$



3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

Sol...

❏ Convert to Compressed Notation

Given IPv6 address: 2001:0db8:abcd:0000:0000:0000:0000/64

👉 Compressed form: 2001:db8:abcd::/64

Explanation:

- Leading zeros in each block are removed.
- Consecutive blocks of zeros are replaced by :: (only once per address).

❏ Expand Back to Full Hexadecimal Representation

👉 Expanded form: 2001:0db8:abcd:0000:0000:0000:0000/64

Explanation:

- Each block is restored to four hexadecimal digits.
- The :: expands to the correct number of zero blocks to make eight total.

❏ Identify the Network Prefix

With a /64 prefix, the network prefix is: 2001:0db8:abcd:0000::/64

Meaning:

- The first 64 bits represent the network portion.
- The last 64 bits represent the host/interface portion.
- This prefix identifies all devices within the same subnet.

❑ Calculate Total Number of Host Addresses in the Subnet

IPv6 addresses are 128 bits long. For a /64 subnet, host bits = $128 - 64 = 64$.

$$2^{64} = 18,446,744,073,709,551,616$$

- ✓ Total host addresses per subnet: 18,446,744,073,709,551,616 (≈ 18.4 quintillion)

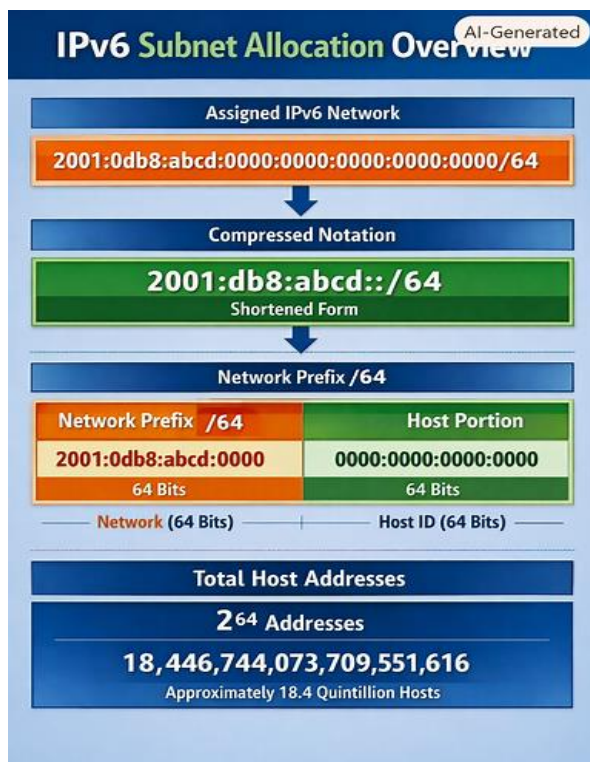
Explanation: Each subnet can support an enormous number of devices — ideal for IoT and medical sensors.

❑ Total Number of Possible Host Addresses in the Network

Since the entire network block is /64, the total number of host addresses available within this network is the same:

$$2^{64} = 18,446,744,073,709,551,616$$

- ✓ Total possible host addresses: 18,446,744,073,709,551,616



4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Sol...

❑ Destination Subnet

The packet's destination IP address is 192.168.2.45. Each routing table entry represents a subnet with a /24 mask (255.255.255.0).

- The /24 mask means the first 24 bits (192.168.2) identify the network. ☒ Therefore, 192.168.2.45 belongs to the 192.168.2.0/24 subnet.

❑ Matching Routing Table Entry

The router compares the destination IP with its routing table entries:

Routing Table Entry	Match?
192.168.1.0/24	 No
192.168.2.0/24	 Yes
192.168.3.0/24	 No

3 Next-Hop Network/Interface

Each routing table entry typically maps to a next-hop IP or interface. Since the packet matches 192.168.2.0/24, it will be forwarded through the next-hop or interface associated with that entry — for example:

- Next-hop: Router interface connected to the 192.168.2.0/24 network
- Interface: e.g., GigabitEthernet0/2 (depending on router configuration)

✓ Forwarding decision: Send the packet via the interface linked to 192.168.2.0/24.

4 Default Route (if no match found)

If the destination IP does not match any routing table entry, the router uses the default route (often called the “gateway of last resort”).

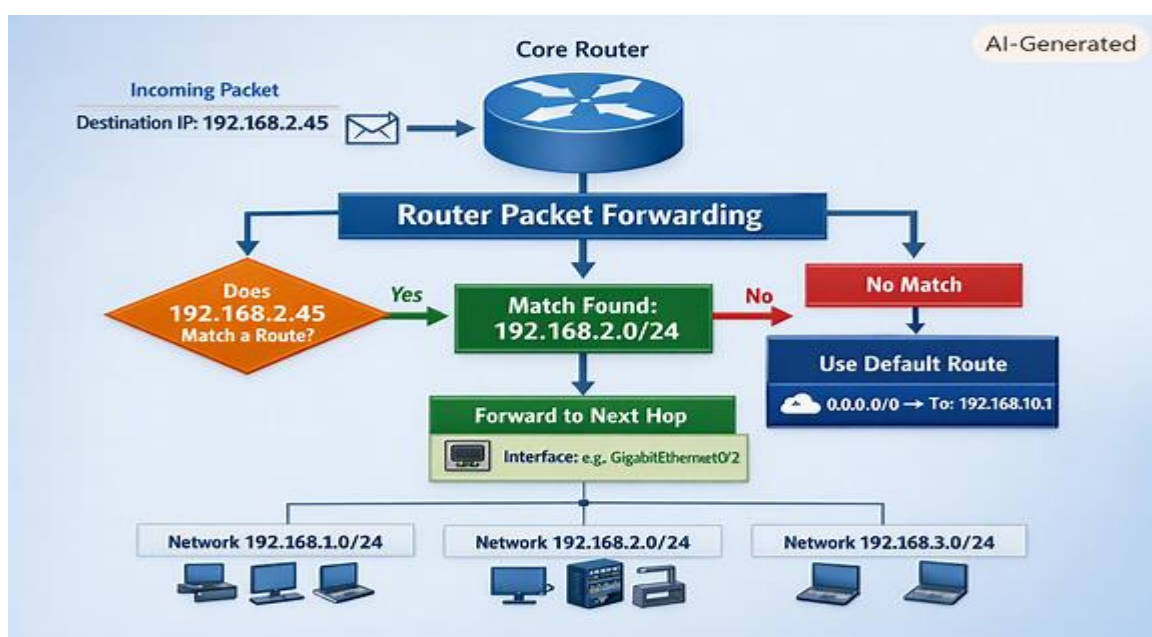
- Default route format: 0.0.0.0/0
- Purpose: Matches all addresses not found in the routing table.
- Example:

Code ^

Copy

```
0.0.0.0/0 → Next-hop 192.168.10.1
```

✓ Default route used: 0.0.0.0/0 (forward to the ISP’s upstream gateway)



5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

Sol..

❑ Network Address and Broadcast Address

Department	Subnet Mask	Network Address	Broadcast Address
Administration LAN	/26	192.168.10.0	192.168.10.63
Finance LAN	/26	192.168.10.64	192.168.10.127

- A /26 subnet has 64 total addresses (from 0–63, 64–127, etc.).
- The first address is the network address, and the last is the broadcast address.

❑ Valid Host IP Address Range

Department	Valid Host Range
Administration LAN	192.168.10.1 – 192.168.10.62
Finance LAN	192.168.10.65 – 192.168.10.126

- Host addresses exclude the network and broadcast addresses.
- Each subnet provides 62 usable host addresses.

📌Suggested IP Addresses for Three Computers

Department	Suggested IPs for Computers
Administration LAN	192.168.10.10, 192.168.10.20, 192.168.10.30
Finance LAN	192.168.10.70, 192.168.10.80, 192.168.10.90

Note:

These are sample host IPs within each valid range. They can be assigned to PCs, printers, or other devices.

📌Default Gateway for Each LAN

Department	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65

- Each LAN's hosts use the router interface IP as their default gateway to communicate outside their subnet.
- The router routes traffic between the two LANs and to external networks.

